

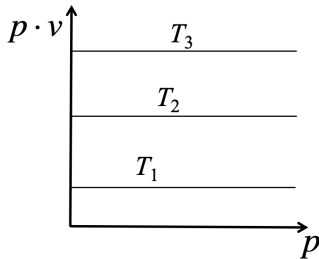
3. Properties of Real Substances described by equations

3.1 The $p \cdot v$, p -diagram and the virial equation of real gases

For an ideal gas with $p \cdot v = R \cdot T$, there is

$$\left(\frac{\partial(p \cdot v)}{\partial p} \right)_T = 0;$$

i.e., in a $p \cdot v$, p -diagram, isotherms are horizontal lines:



In reality, there are forces of attraction and repulsion between molecules.

- Attraction: the real $p \cdot v$ is smaller than that for an ideal gas (at given p and T , v is smaller)
- Repulsion: the real $p \cdot v$ is bigger than that for an ideal gas (at given p and T , v is larger)

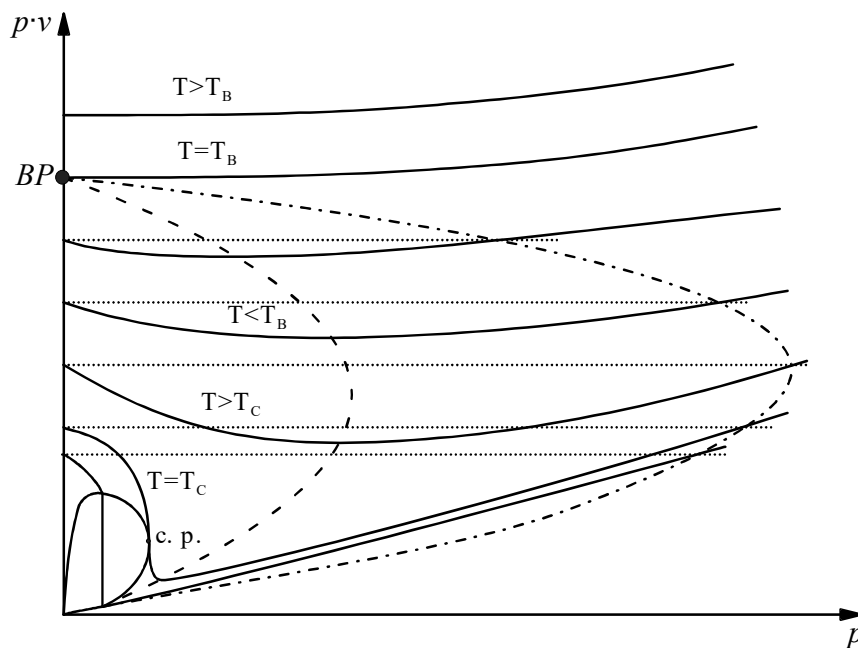


Figure: $p \cdot v$, p -diagram of a real substance (qualitative)

- - - - Classic Ideal Curve

- - -: Boyle-Curve

Classical ideal-curve: curve on which $p \cdot v = R \cdot T$ for the determination of p or v or T at $p > 0$.

T_B = Boyle-temperature, c.p. = critical point.

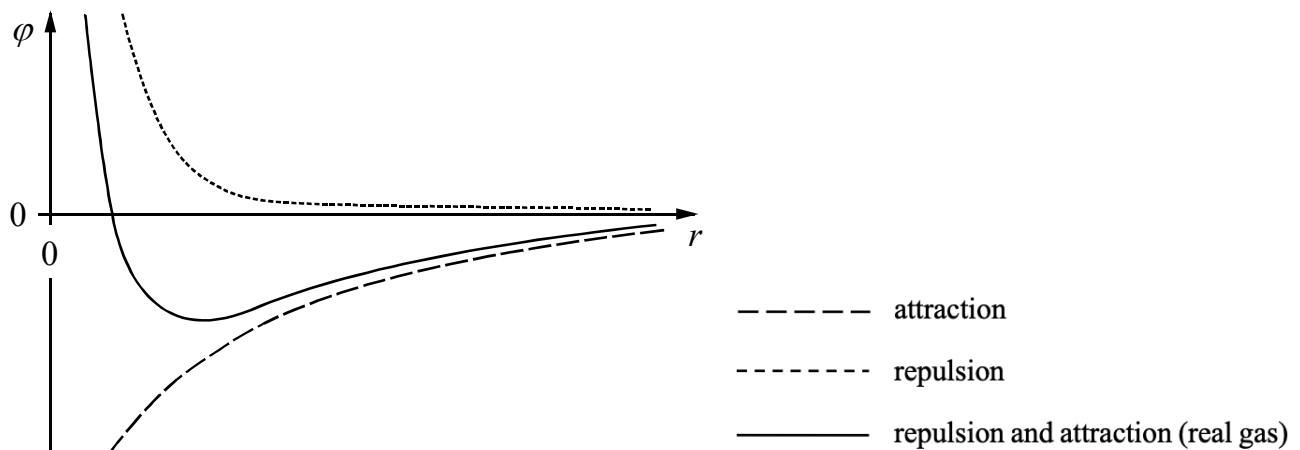
Further consequences from $p \cdot v = R \cdot T$ are generally not valid.

Boyle-curve: curve on which $\left(\frac{\partial(p \cdot v)}{\partial p} \right)_T = 0$ holds.

BP = Boyle point: $\left(\frac{\partial(p \cdot v)}{\partial p} \right)_{T=T_B} = 0$ (at $p \rightarrow 0$)

For $p \rightarrow 0$, the behavior of the real gas does not become identical to the behavior of the ideal gas of the same molar mass!

Ideal curves in general: Place where the real gas has at least one property in common with the ideal gas of the same molar mass.



Virial equations

Taylor series for the isotherms in a $p \cdot v, p$ -diagram developed at $p = 0$ or in a $p \cdot v, \rho$ -diagram, developed at $\rho = 0$:

“technical” form
$$p \cdot v = R \cdot T + B^*(T) \cdot p + C^*(T) \cdot p^2 + D^*(T) \cdot p^3 + \dots \quad (9.18)$$

“physical” form
$$p \cdot v = R \cdot T \cdot (1 + B(T) \cdot \rho + C(T) \cdot \rho^2 + D(T) \cdot \rho^3 + \dots) \quad (9.19)$$

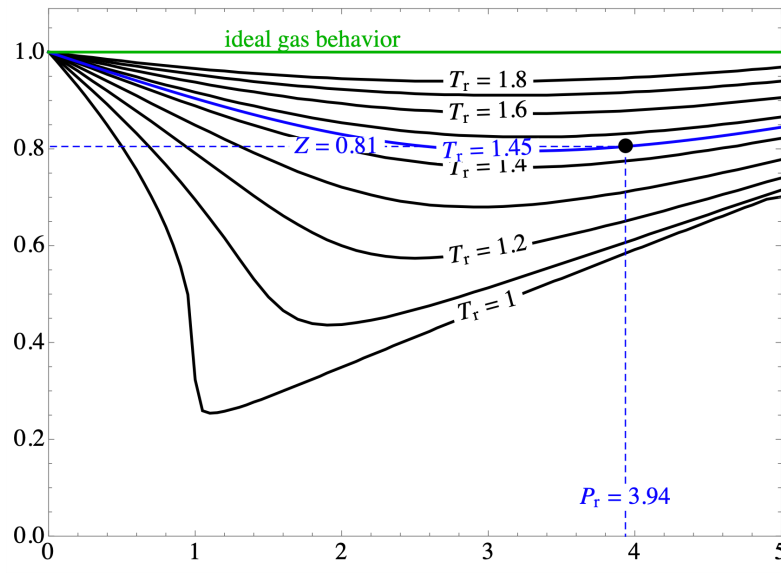
$$(1 + B(T) \cdot \rho + C(T) \cdot \rho^2 + D(T) \cdot \rho^3 + \dots) = Z$$

For very small p or ρ , the characteristics are determined by B^* or B .

Compressibility factor $Z = \frac{p \cdot v}{R \cdot T}$:

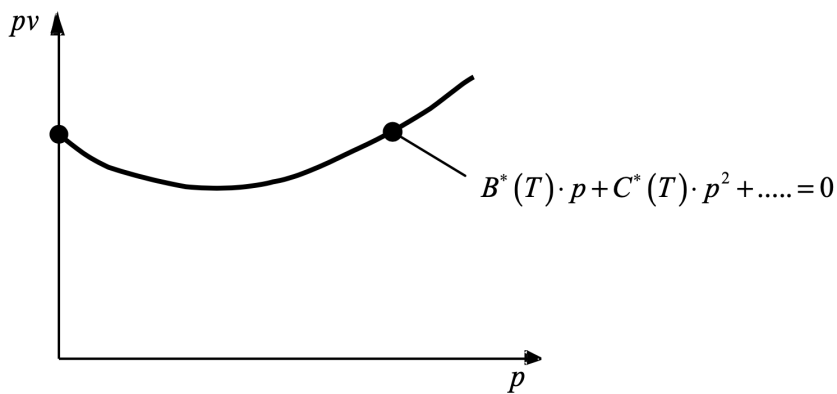
Examples:

	T = 300 K			100 K
	He	Ar	Ne	N ₂
p = 0.1 MPa	1.0005	0.9999	0.9998	0.990
5 MPa	1.024	0.971	0.996	0.243
100 MPa	1.439	1.675	1.99	3.89



Quelle: <https://demonstrations.wolfram.com/CompressibilityFactorCharts/>

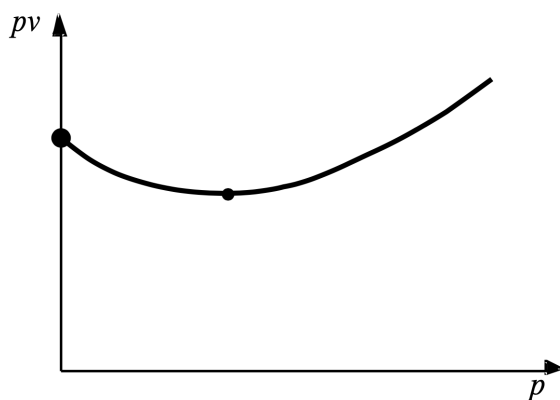
For which p does $pv=RT$ hold?



$B^*(T) \cdot p + C^*(T) \cdot p^2 + D^*(T) \cdot p^3 + \dots = 0 \quad \Rightarrow \quad$ leads to the classical ideal curve

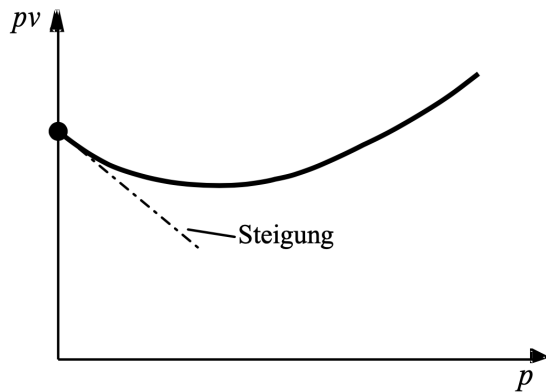
Where is the minimum of the isotherm?

$$\left(\frac{\partial pv}{\partial p} \right)_T = 0 \Rightarrow B^*(T) + 2C^*(T) \cdot p + 3D^*(T) \cdot p^2 + \dots = 0$$



What is the slope of the curve at $p = 0$?

$$\left(\frac{\partial pv}{\partial p}\right)_T = B^*(T) + 2C^*(T) \cdot p + \dots, \quad \lim_{p \rightarrow 0} \left(\frac{\partial pv}{\partial p}\right)_T = B^*(T)$$

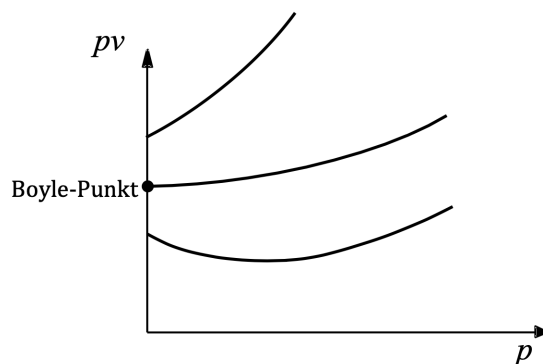


(Steigung = Slope; can be negative or positive).

Are there isotherms which have no minimum? (Repulsion always bigger than attraction.)

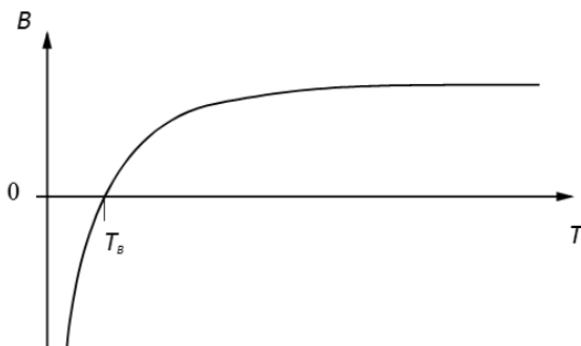
$\Rightarrow$ Minimum would be at $p = 0$

$$\left(\frac{\partial pv}{\partial p}\right)_T \Big|_{p=0} = B^*(T_B)$$



Remark: $B(T)$ can be determined experimentally with an uncertainty between 0.1 % and 1.0 %. The differences between the experimental values of $C(T)$ in the literature are between 1 % and 10 %.

Figure: Qualitative dependence of $B(T)$:



How are B and B^* related?

$$B = \frac{1}{R \cdot T} \cdot \lim_{p \rightarrow 0} \left(\frac{\partial(p \cdot v)}{\partial p}\right)_T \quad \text{and} \quad R \cdot T \cdot \lim_{p \rightarrow 0} \left(\frac{\partial p}{\partial \rho}\right)_T = 1 \quad \text{yields:} \quad B = \lim_{p \rightarrow 0} \left(\frac{\partial(p \cdot v)}{\partial \rho}\right)_T$$

$$B^* = \lim_{\rho \rightarrow 0} \left(\frac{\partial(p \cdot v)}{\partial \rho} \right)_T$$

$p \rightarrow 0$ corresponds to $\rho \rightarrow 0$ (real gas at $T \neq 0$), therefore: $B = B^*$

Exercise: Show that $C = R T C^*$ and $D = (R T)^2 D^*$

3.2 Modifications of the Ideal Gas as Models for Real Gases

3.2.1 Mass Points with Attraction

Formulation: $p \cdot v + \frac{a}{v} = R \cdot T$, $a = \text{constant}$

The influence of $\frac{a}{v}$ vanishes if $v \rightarrow \infty$. There follows

$\lim_{p \rightarrow 0} p \cdot v = R \cdot T$. This is in partial agreement with the behavior of a real gas.

Equation of state:

$$\underbrace{\left(p + \frac{a}{v^2} \right)}_{\hat{=} p_{id.}} \cdot v = R \cdot T.$$

The forces of attraction, therefore, diminish the pressure by a/v^2 compared with p_{id} of an ideal gas of the same molecular mass.

$$p = p_{id} - \frac{a}{v^2} \quad \frac{a}{v^2} = \text{“cohesive pressure”}$$

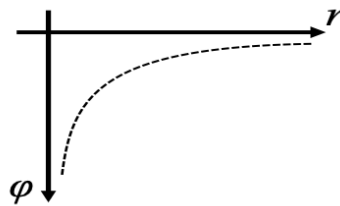


Figure: Interaction potential of two molecules of a rigid sphere gas as a model of a gas with attractive interaction.

3.2.2 Molecules with only Repulsion

Simplest model: rigid spheres (hard sphere molecules):

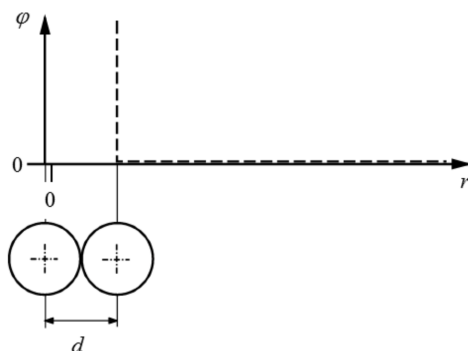


Figure: Potential φ of the interaction of two rigid spheres. Model of molecules with only repulsion.

For the formulation of the equation of state, we consider the volume B which is not accessible for the centers of the molecules. (The whole space is accessible for mass points.) The volume of the gas

consisting of rigid spheres is larger by B than the volume of an ideal gas (=mass points without attraction).

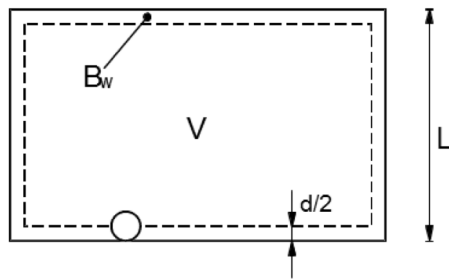
Relative to the mass: $b = \frac{B}{m}$

$v = v_{id} + b$ and therefore $v_{id} = v - b$

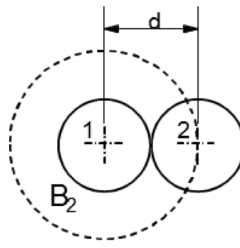
Equation of state of the gas consisting of rigid spheres in analogy to the equation of state of the ideal gas of the same molecular mass:

$$p \cdot (v - b) = R \cdot T$$

Relation between B and the volume of the rigid spheres:



a.



b.

Space, not accessible for the centers of the rigid spheres

a. along the wall: B_W

b. space B_2 not accessible for the center of the sphere 2 during the collision of the spheres 1 and 2.

We first assume that the sphere 1 can move freely.

a: Collision with the wall:

$V \sim L^3$ L = linear dimension of the container

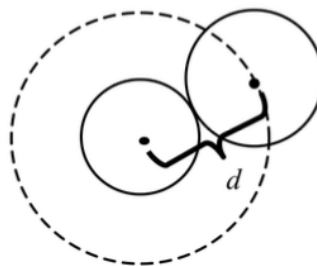
$$B_W \sim d \cdot L^2$$

It follows that $\frac{B_W}{V} \sim \frac{d}{L} \ll 1 \Rightarrow$ we neglect B_W .

b: Collision of two spheres:

Preliminary assumption: Particle 1 can move freely. Then, for the center of particle 2 a spherical volume of radius d is obstructed; that is, a volume

$$B_2 = \frac{4}{3} \pi \cdot d^3$$



Intrinsic volume of a sphere

$$V_S = \frac{\pi}{6} d^3 \Rightarrow \frac{B_2}{V_S} = 8$$

On average, for each particle the volume

$$B = 4 \cdot V_S$$

is obstructed (consideration of particle pairs induces a factor of 1/2).

Per mass of the sphere (μ = mass of a particle)

$$b = \frac{4}{\mu} \cdot V_S$$

Example N_2 : $d \approx 3 \text{ \AA} = 3 \cdot 10^{-10} \text{ m}$ $b \approx 10^{-3} \text{ m}^3/\text{kg}$.

for comparison: at 300 K, there is $v \approx 0.8 \text{ m}^3/\text{kg}$

3.2.3 van der Waals-Gas

Gas with attraction according to $\left(p + \frac{a}{v^2}\right)$ and rigid volume of the molecules according to $(v - b)$

Equation of state:

$$\left(p + \frac{a}{v^2}\right) \cdot (v - b) = R \cdot T$$

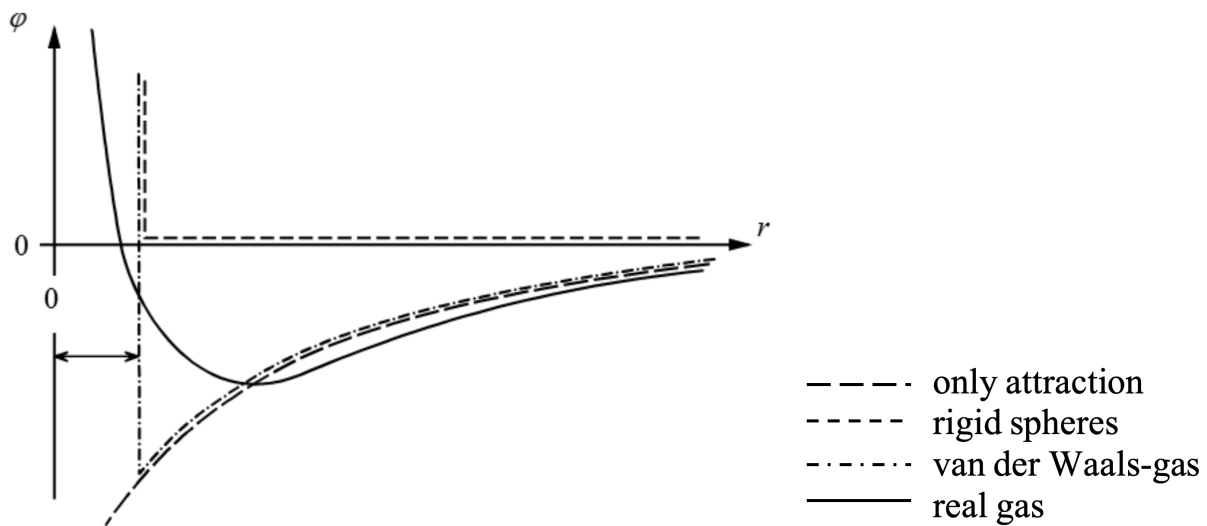


Figure: Interaction potential of the two molecules.

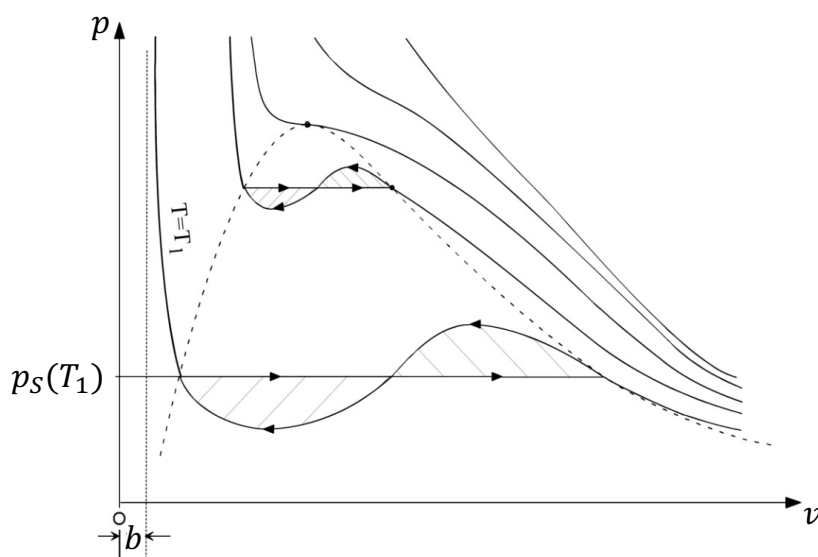


Figure: p, v -diagram of the van der Waals-gas

$p_S(T)$ is determined using the *Maxwell criterion* $\oint p \cdot dv = 0$. It is based on the statement that the cycle on the horizontal isotherm $T_S(p)$ and the curved isotherm of the van der Waals equation must not produce any work, because overall $T = \text{const}$.

3.2.4 Examples for consequences from the Equations of State

We want: Volume dependence of internal energy $\left(\frac{\partial u}{\partial v}\right)_T$ and pressure dependence of enthalpy $\left(\frac{\partial h}{\partial p}\right)_T$ for the different model gases.

There is valid generally that $\left(\frac{\partial u}{\partial v}\right)_T = T \left(\frac{\partial p}{\partial T}\right)_v - p$ and $\left(\frac{\partial h}{\partial p}\right)_T = v - T \left(\frac{\partial v}{\partial T}\right)_p$.

van der Waals-gas: $\left(p + \frac{a}{v^2}\right) \cdot (v - b) = R \cdot T$

$\left(\frac{\partial u}{\partial v}\right)_T$ for a van der Waals-gas:

$$\left. \begin{aligned} p &= \frac{R \cdot T}{v - b} - \frac{a}{v^2} \\ T \cdot \left(\frac{\partial p}{\partial T}\right)_v &= \frac{R \cdot T}{v - b} \end{aligned} \right\} \left(\frac{\partial u}{\partial v}\right)_T = \frac{a}{v^2}$$

$\left(\frac{\partial h}{\partial p}\right)_T$ for a van der Waals-gas:

First approximation: $p v - p b + \frac{a}{v} - \frac{ab}{v^2} \approx p \cdot v - p b + \frac{a}{v}$

Second approximation: $\frac{a}{v} \approx \frac{ap}{RT}$

$$\left. \begin{aligned} p \cdot v + \frac{ap}{RT} - p b &= R T \\ v &= \frac{RT}{p} + b - \frac{a}{RT} \\ T \cdot \left(\frac{\partial v}{\partial T}\right)_p &\approx \frac{R \cdot T}{p} + \frac{a}{RT} \end{aligned} \right\} \left(\frac{\partial h}{\partial p}\right)_T \approx b - \frac{2a}{RT}$$

Real gas described by the virial equation:

$$\text{“physical” form: } p \cdot v = R \cdot T \cdot (1 + B \cdot \rho + C \cdot \rho^2 + \dots)$$

$$\text{“technical” form: } p \cdot v = R \cdot T + B^* \cdot p + C^* \cdot p^2 + \dots$$

The shortened forms for large v and small p are applied in the following:

It follows from $p \cdot v = R \cdot T \cdot (1 + B \cdot \rho)$

that $p = \frac{R \cdot T}{v} + \frac{B \cdot R \cdot T}{v^2}$

and $T \cdot \left(\frac{\partial p}{\partial T}\right)_v = \frac{R \cdot T}{v} + \frac{B \cdot R \cdot T}{v^2} + \frac{R \cdot T^2}{v^2} \cdot \frac{dB}{dT}$

and therefore $\left(\frac{\partial u}{\partial v}\right)_T = \frac{R \cdot T^2}{v^2} \cdot \frac{dB}{dT}$

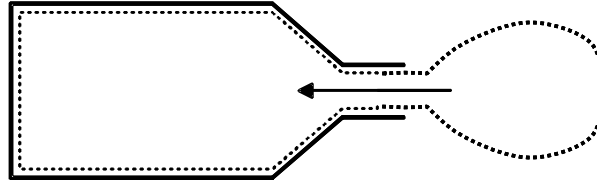
$p v = RT + B^* \cdot p$ yields:

$$v = \frac{RT}{p} + B^* T \left(\frac{\partial v}{\partial T} \right)_p = \frac{RT}{p} + T \cdot \frac{dB^*}{dT} \left(\frac{\partial v}{\partial T} \right)_p = \frac{RT}{p} + T \cdot \frac{dB^*}{dT} \left\{ \left(\frac{\partial h}{\partial p} \right)_T = B^* - T \cdot \frac{dB^*}{dT} \right.$$

3.3 Joule-Thomson-Effect (“Throttling Effect”)

3.3.1 Preliminary considerations on the Joule-Thomson Effect

Experiment from thermodynamics I – tutorial

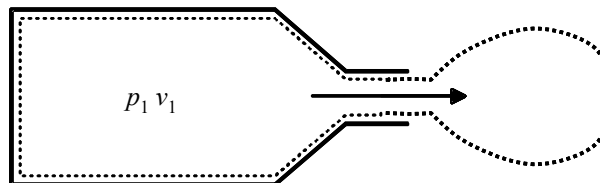


$$\Delta u = p_a \cdot v_a = R \cdot T_a$$

Perfect gas: $c_v(T - T_a) = RT_a$, $T = \frac{c_v + R}{c_v} \cdot T_a = \frac{c_p}{c_v} \cdot T_a$

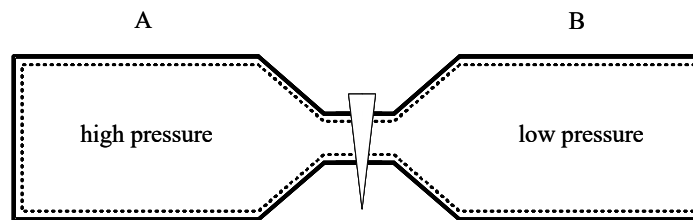
$\Rightarrow$ temperature rises, atmosphere does work.

“Inverse” case:



Gas does work of displacement $\Rightarrow$ temperature decreases.

Third case:



After opening the valve: Temperature of A decreases, temperature of B increases.

After temperature equilibration: Same temperature as before in the case of an ideal gas.

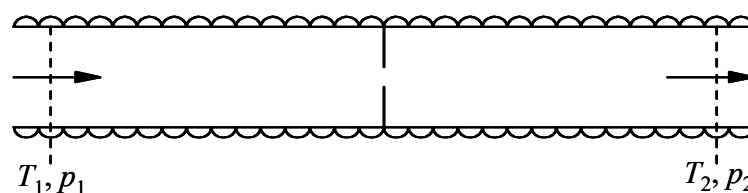
3.3.2 Integral Joule-Thomson-Effect

The integral Joule-Thomson-effect is the temperature difference caused by a pressure difference at unchanged enthalpy. We can formulate it as

Integral Joule-Thomson-Effect (J-T-E):

$$\frac{T_2 - T_1}{p_2 - p_1} \quad \text{with} \quad h_2(T_2, p_2) = h_1(T_1, p_1).$$

The integral J-T-E can be realized by adiabatic throttling with $w = 0$, $\Delta \frac{c^2}{2} = 0$, $\Delta z = 0$:



3.3.3 Differential Joule-Thomson-Effect

$$\left(\frac{\partial T}{\partial p}\right)_h \equiv \mu$$

It follows from $dh = c_p \cdot dT + \left(\frac{\partial h}{\partial p}\right)_T \cdot dp$ and $dh = 0$ that

$$\mu = -\frac{1}{c_p} \cdot \left(\frac{\partial h}{\partial p}\right)_T \quad \mu = 0 \text{ for an ideal gas, since } \left(\frac{\partial h}{\partial p}\right)_T = 0$$

and in general, using $\left(\frac{\partial h}{\partial p}\right)_T = v - T \cdot \left(\frac{\partial v}{\partial T}\right)_p$:

$$\mu = -\frac{1}{c_p} \cdot \left(v - T \cdot \left(\frac{\partial v}{\partial T}\right)_p \right)$$

The term $\left(\frac{\partial h}{\partial p}\right)_T$ is often called *isothermal throttling effect* and is abbreviated as Φ .

For the van der Waals gas:

$$\left(\frac{\partial h}{\partial p}\right)_T = b - \frac{2a}{RT}$$

- $\left(\frac{\partial h}{\partial p}\right)_T > 0 \Rightarrow \mu < 0 \Rightarrow p \downarrow \Rightarrow T \uparrow$
- $\left(\frac{\partial h}{\partial p}\right)_T < 0 \Rightarrow \mu > 0 \Rightarrow p \downarrow \Rightarrow T \downarrow$

3.3.4 Inversion Curves of the Joule-Thomson-Effect

The inversion curve of the differential Joule-Thomson-effect is formed by the points with $\mu = 0$. These are the points, e.g., in a p, h -diagram, in which the isotherms have a vertical tangent.

$$\left(\frac{\partial p}{\partial h}\right)_T = \infty \Rightarrow \left(\frac{\partial h}{\partial p}\right)_T = 0 \Rightarrow \mu = 0$$

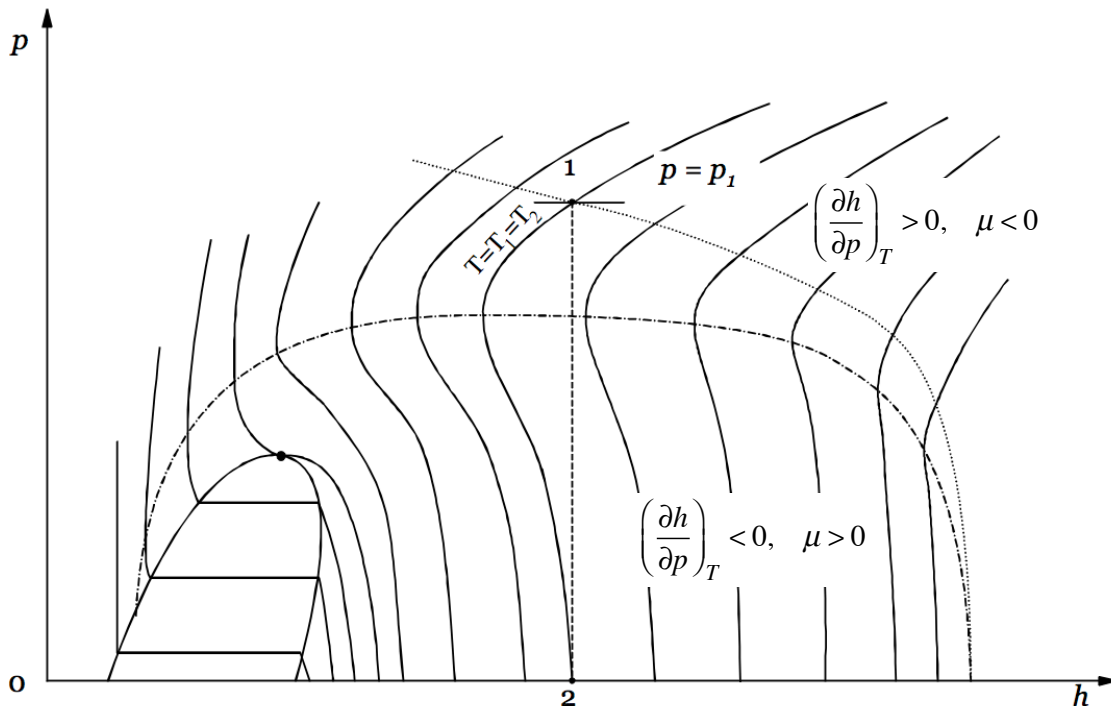


Figure:
 - - - Inversion curve of the differential Joule-Thomson-effect. The temperature decreases with a decrease of the pressure at $h = \text{const.}$ below this curve: μ is positive. The temperature increases outside this area: μ is negative.
 Inversion curve of the integral Joule-Thomson-effect.

An inversion curve of the integral Joule-Thomson-effect can be introduced if the final pressure p_2 is fixed. This curve then is formed by the points T_1, p_1 for which $T_2 = T_1$ with $h_2(T_2, p_2) = h_1(T_1, p_1)$. Such a point T_1, p_1 is shown in figure 9.14 for $p_2 = 0$.

Some substances and their inversion temperatures: H_2 : -80°C , He : -238°C , N_2 : 577°C , O_2 : 767°C

3.4 Classical Principle of Corresponding States

Until now we considered the properties of real substances mainly in qualitative diagrams. The diagrams were dominated by the critical point. We get quantitative diagrams for individual substances if we take the individual critical properties as scale units:

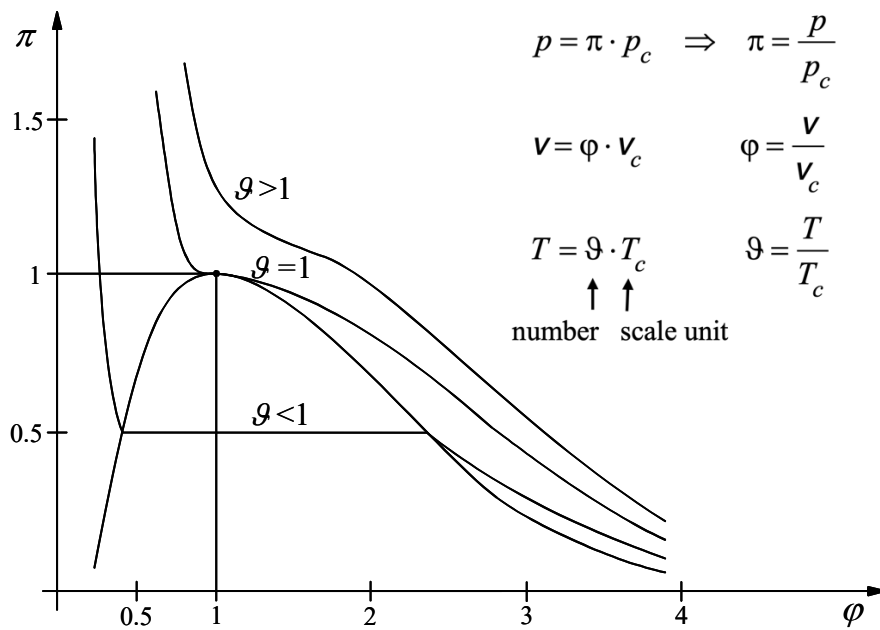


Figure: Dimensionless pressure, volume-diagram (generalized diagram). Critical properties as units.

The numerical values π, φ, ϑ are related (“normalized”) to the critical state variables.

The classical principle of corresponding states is the attempt to transfer the arguments which we based on the pressure, volume-diagram to the equation of state: i.e., to fix the surface $p(v, T)$ at the critical point and to generalize it as $\pi(\varphi, \vartheta)$. “Corresponding” states $\pi_1, \varphi_1, \vartheta_1$ then could be transformed into individual states p_1, v_1, T_1 by means of the critical data p_c, v_c, T_c .

As an example, we fix the van der Waals-equation to the critical point:

The three constants a, b, R are determined by means of the following three conditions at the critical point. The conditions correspond to the horizontal tangent at the point of inflexion of the critical isotherm in the p, v -diagram.

It is valid on the critical isotherm $T = T_c$ at $v = v_c$ that

1. $p = p_c$
2. $\left(\frac{\partial p}{\partial v}\right)_T = 0$
3. $\left(\frac{\partial^2 p}{\partial v^2}\right)_T = 0$

The van der Waals-equation

$\left(p + \frac{a}{v^2}\right) \cdot (v - b) = R \cdot T$ can be written in the form

$$v^3 - \left(b + \frac{R T}{p}\right) v^2 + \frac{a}{p} v - \frac{a b}{p} = 0$$

$$\left(p + \frac{a}{v^2}\right) (v - b) = R T$$

$$\pi = \frac{p}{p_k}$$

$$\varphi = \frac{v}{v_k}$$

$$\vartheta = \frac{T}{T_k}$$

$$\left(p_k \cdot \pi + \frac{a}{v_k^2 \varphi^2}\right) (v_k \varphi - b) = R \cdot T_k \cdot \vartheta$$

$$\left(\pi + \frac{a}{v_k^2 p_k \varphi^2}\right) \left(\varphi - \frac{b}{v_k}\right) = \frac{R \cdot T_k}{p_k \cdot v_k} \cdot \vartheta$$

$$\left(\pi + \frac{A}{\varphi^2}\right) (\varphi - B) = C \cdot \vartheta$$

$$A = \frac{a}{v_k^2 p_k}$$

$$B = \frac{b}{v_k}$$

$$C = \frac{R \cdot T_k}{p_k \cdot v_k}$$

Solve for π : $\pi = \frac{C \cdot \vartheta}{\varphi - B} - \frac{A}{\varphi^2}$

The volume is contained in the third degree. We now develop the critical isotherm

$p = p(v, T = T_c)$ at $v = v_c$ into a Taylor-series till the term of third degree.

$$p = p_c + \left(\frac{\partial p}{\partial v}\right)_T (T = T_c, v = v_c) \cdot (v - v_c) + \frac{1}{2} \cdot \left(\frac{\partial^2 p}{\partial v^2}\right)_T (T = T_c, v = v_c) \cdot (v - v_c)^2 + \frac{1}{6} \cdot \left(\frac{\partial^3 p}{\partial v^3}\right)_T (T = T_c, v = v_c) \cdot (v - v_c)^3$$

The three conditions are fulfilled by introducing them into the series. We get

$$(v - v_c)^3 = 0$$

$$v^3 - 3v_c v^2 + 3v_c^2 v - v_c^3 = 0$$

Equation 9.40a can be interpreted in the following way:

The volume becomes threefold v_c . The three points of intersection of an isotherm

$T_1 < T_c$ with the isobar $p_s(T_1)$ form the critical point if $T_1 \rightarrow T_c$.

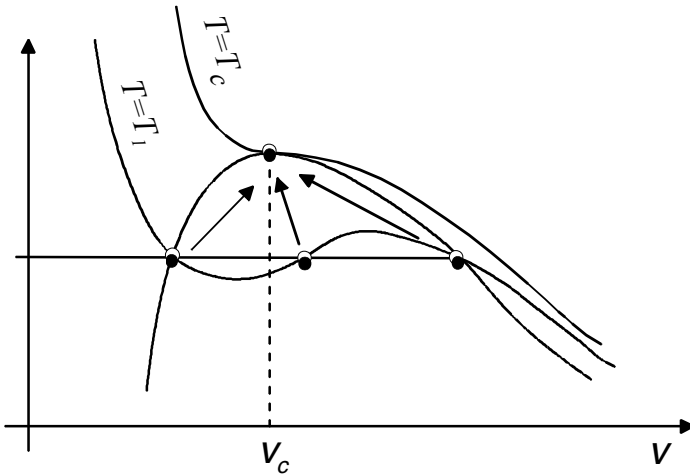


Figure: Concurrence of the three intersection points of $T = T_1$ with $p_s(T_1)$ at the critical point.

At the critical point, there hold several conditions:

1. Point fulfils the equation ($\pi = 1, \varphi = 1, \vartheta = 1$)
2. At $\pi = 1, \varphi = 1, \vartheta = 1$, there is a horizontal inflection tangent

Determining the Derivatives:

$$\begin{aligned} \left(\frac{\partial \pi}{\partial \varphi}\right)_{\vartheta} &= -\frac{C \cdot \vartheta}{(\varphi - B)^2} + \frac{2A}{\varphi^3} & \left(\frac{\partial \pi}{\partial \varphi}\right)_{\vartheta=1} &= -\frac{C}{(1-B)^2} + 2A = 0 \\ \left(\frac{\partial^2 \pi}{\partial \varphi^2}\right)_{\vartheta} &= 2 \frac{C \cdot \vartheta}{(\varphi - B)^3} - \frac{6A}{\varphi^3} & \left(\frac{\partial^2 \pi}{\partial \varphi^2}\right)_{\vartheta=1} &= \frac{2C}{(1-B)^3} - 6A = 0 \end{aligned}$$

Additionally: $(1 + A)(1 - B) = C$ $1 = \frac{C}{1-B} - A$

We get 3 equations for 3 unknowns:

$$A = \frac{C}{2(1-B)^2} \quad A = \frac{C}{3(1-B)^3} \quad A = \frac{C}{1-B} - 1$$

With the solution $A = 3, B = \frac{1}{3}, C = \frac{8}{3}$

$$\left(\pi + \frac{3}{\varphi^2}\right) \cdot \left(\varphi - \frac{1}{3}\right) = \frac{8}{3} \cdot \vartheta$$

This would be the generalized equation of state of the classical correspondence principle, built up from dimensionless quantities, which would result in individual equations of state with individual critical data as units of measurement.

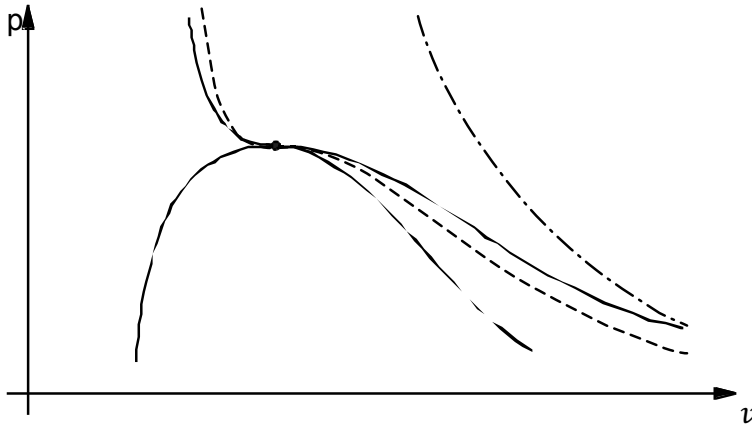


Figure: Difference between the critical Isotherm of the Van der Waals-Gas and the critical isotherm of a real substance: — real substance, --- van der Waals-gas, -·- ideal gas

Reasons for the inaccuracy of the classical principle of corresponding states are that with only three conditions to adapt to the critical point, not enough parameters are available to reflect the individual behaviour of real substances.

Such parameters would be:

- the molar mass M

and (based on the interaction potential between two molecules):

- the depth ε of the well of the potential
- the distance $\sigma = r(\varphi = 0)$
- the slope $\frac{d\varphi}{dr}(\varphi = 0)$

Deviations of the potential from spherical symmetry are still not regarded as well as the interaction between three or more molecules.

One tried to develop extended principles of corresponding states which contain more than three individual parameters and searched for additional individual quantities, characteristic for substances to fix the equations to such quantities. Many equations were formulated. The fast development of computers has led the efforts into another direction: Bulky formulations – often with more than 100 constants – are possible now. The most recent formulations are stored electronically and can be used easily. But such formulations offer also the possibility to display the available data sets with all their uncertainties. „With sixty constants, you can fit the surface of a cow,“ Lester Haar, 1923-1990, National Institute of Standards and Technology, Gaithersburg, Maryland, U.S.A. L. Haar has developed formulations for the thermodynamic properties e.g., of NH_3 and H_2O . He had to base them on many sets of more or less accurate experimental data. To estimate and take into account the reliability of such sets is an important and time-consuming task. Additionally, the formulations have to be consistent with the general relations between the thermodynamic properties in all details. By computers, the work also in this field became more efficient but not less laborious. The possibilities which computers offer make reliable experimental data especially important.

3.5 Metastable States

Metastable states are one-phase states of limited stability. Systems can go from meta-stable states into absolutely stable two-phase states after a disturbance. The van der Waals-equation as an equation of third order in v describes also metastable states although its structure is very simple and the ideas on which it is based are very simple (e. g. rigid spheres):

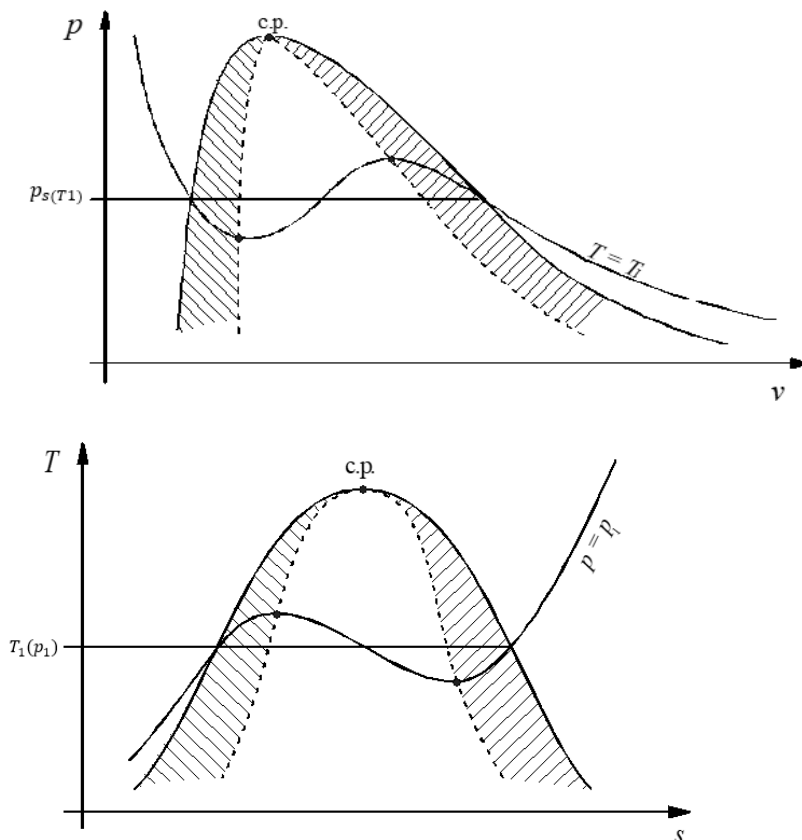


Figure: Metastable states

undercooled vapor: $T < T_s(p)$ and $p > p_s(T)$, region

superheated liquid: $T > T_s(p)$ and $p < p_s(T)$, region



A system can reach metastable one-phase states on the curved isotherms or isobars from the region of absolute stability outside the two-phase region without decaying into coexisting two phases.

Temperature and pressure are unambiguously connected only for a two-phase system: This is described by

- the vapor pressure curve;
- the horizontal isotherms in the p, v -diagram and the horizontal isobars in the T, s -diagram on which the center of gravity of the two partial systems lies: liquid on the left boundary, vapor on the right boundary of the two-phase region.

A system is absolutely unstable between the two regions of metastability, between the superheated liquid and the undercooled vapor. There would be:

- in the p, v -diagram: $\left(\frac{\partial p}{\partial v}\right)_T > 0$, which is impossible out of general demands of stability

and

- in the T, s -diagram: $\left(\frac{\partial T}{\partial s}\right)_p = \frac{T}{c_p} < 0$, which is impossible because T and c_p never are negative.

There are valid for the h, s -diagram with $dh = T \cdot ds + v \cdot dp$

$$\left(\frac{\partial h}{\partial s}\right)_p = T \quad \text{and} \quad \left(\frac{\partial^2 h}{\partial s^2}\right)_p = \left(\frac{\partial T}{\partial s}\right)_p = \frac{T}{c_p} > 0$$

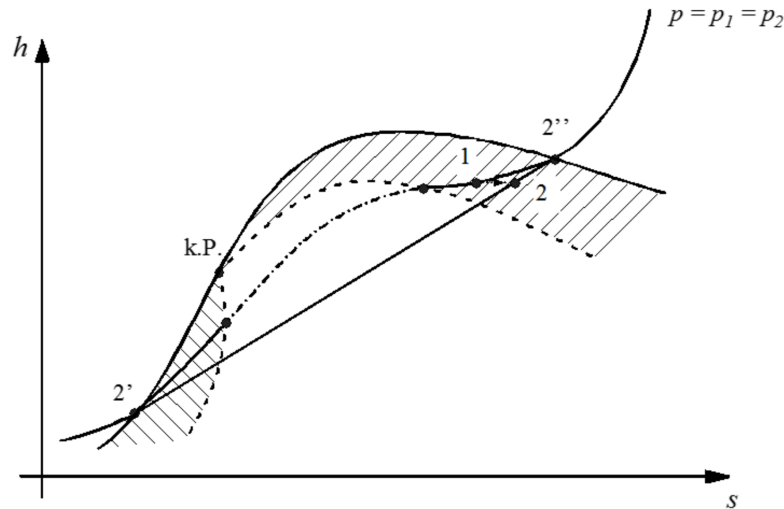


Figure: Increase (by production) of entropy during the irreversibly adiabatic decay of a metastable one-phase system 1 into a stable two-phase system 2.

In the h, s -diagram one can see clearly how the entropy rises during the (adiabatic) decay of a metastable vapor 1 into coexisting liquid 2' and vapor 2'': $s_2 > s_1$. Till now, the decay was considered as isobaric and isenthalpic. It also could be considered at constant volume and at constant internal energy.

In both cases (p and h constant or v and u constant), there would result from the equations $dh = T \cdot ds + v \cdot dp$ or $du = T \cdot ds - p \cdot dv$ that the entropy is not changed during the decay of the metastable system. But, the increase of entropy during the (adiabatic and irreversible) decaying of the metastable system into coexisting phases is realistic because the two differential forms describe only the metastable one-phase system or the absolutely stable two-phase system but not the irreversible transition. With the two differential forms one can describe reversible substitutes for the irreversible

transition by going in the metastable region to the boundary of the two-phase region and from there to the coexisting phases in the two-phase region:

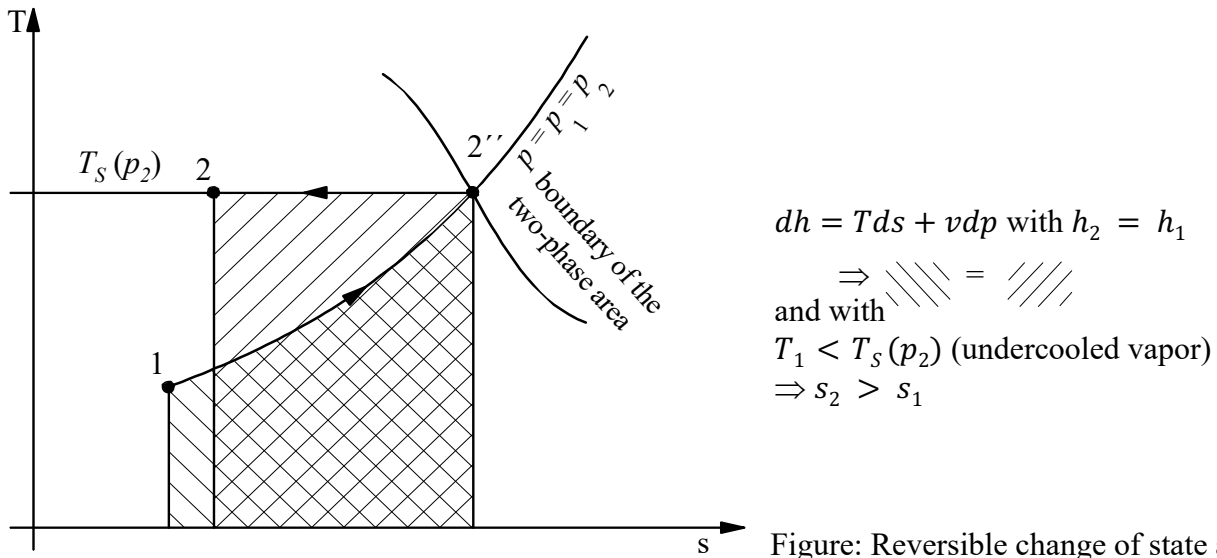
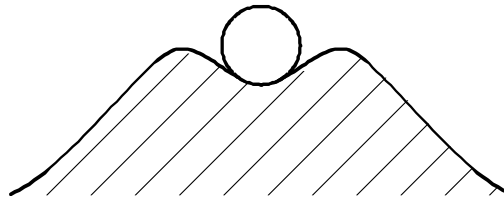


Figure: Reversible change of state as

substitute for the irreversible adiabatic (isenthalpic) transition from a metastable one-phase state 1 to a stable two-phase state 2.

The mechanism which triggers decay of a metastable one-phase system into an absolutely stable two-phase system can be a mechanical impact causing spatial differences of density which grow to the larger differences between the densities of the two phases. But microscopic fluctuations of density due to statistical fluctuations caused by the movement of the molecules may also trigger a decay. Such fluctuations need not lead to bubbles in the metastable liquid or to droplets in the metastable vapor, they can also decay again. This is analogous to a ball rolling back into the position of limited stability in a depression on a summit. The restoring force to the lower level corresponds in the metastable system to a restoring *force* to the higher entropy.



This is no contradiction to $s_2 > s_1$ of the decay into two phases. As long as e.g., the isobars in a h, s -diagram have a positive second derivative, $\left(\frac{\partial^2 h}{\partial s^2}\right)_p > 0$, there will be caused a total state 2 of a lower entropy by a fluctuation of the density (decay into the states 2* and 2**).

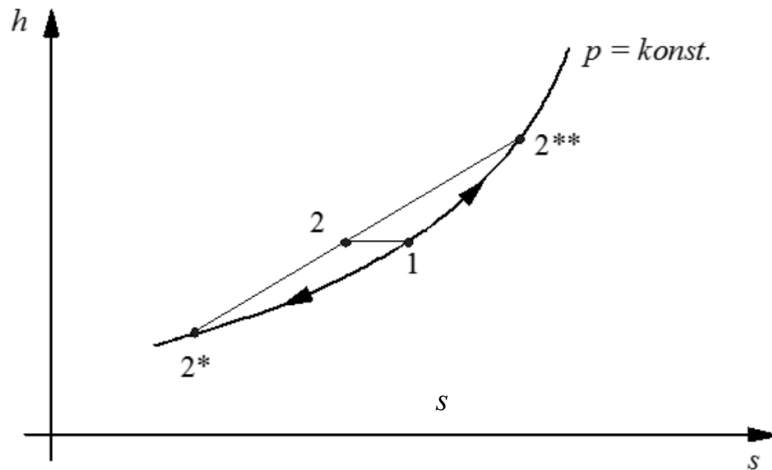


Figure: Decrease of entropy as a restoring force during an isobaric fluctuation of the density as long as $(\partial^2 h / \partial s^2)_p > 0$.

The state 2* of the partial system can move to the left beyond the point of inflection of the isobar if the state 1 of a sub-cooled vapor approaches the point of inflection, i.e., the limit to absolute instability. The system then will decay with rising entropy into a two-phase system. The entropy of this two-phase system reaches a maximum when the total state (in the sense of the lever rule) lies on a straight line which is tangent to the isobar in the two states of the co-existing liquid and gas (2' and 2"); cf. fig. 9.19). The result is also a steady inclination $(\partial h / \partial s)_p = T$ of the isobar.

Remarks:

1. There follows immediately a positive inclination of the isobars in the T, s -diagram from $(\partial^2 h / \partial s^2)_p = (\partial T / \partial s)_p = T / c_p > 0$. The condition $(\partial p / \partial v)_T < 0$ in the p, v -diagram for the stability of a system can be understood easily because with $(\partial p / \partial v)_T > 0$ the restoring forces would be negative: The partial system of which the volume is diminished by a fluctuation would have a diminished pressure, the partial system of the enlarged volume would have an enlarged pressure. The pressure difference will cause a compression of the already compressed partial system and an expansion of the already expanded partial system until the coexisting phases liquid and vapor are reached.
2. Fluctuations of the density can lead to the spontaneous decay of a homogeneous metastable one-phase system into a stable two-phase system. A metastable system can also decay because the second phase is already present in form of particles of other substances:
 - in form of bubbles of unsolved gases in a metastable liquid, free floating or caught in the roughness of containing walls. Such bubbles must be expanded by the vapor against the surface tension. This demands the more superheating above the (equilibrium-) boiling temperature the smaller the bubbles are.
 - in form of aerosol particles in subcooled vapor, e.g., in steam which was subcooled in a fast expansion.

The *heterogeneous* condensation which is triggered by the aerosol particles is mainly not sufficient to reach phase equilibrium during the fast expansion in a steam turbine. Then most of the condensation takes place as *homogeneous* condensation at nuclei which are formed by spontaneous agglomeration of single molecules. We shall not deal with the dependence between the rate of formation of such nuclei (number per time) and the degree of subcooling of the vapor.

3. Metastable states are applied in elementary particle physics:

The path of elementary particles in a subcooled vapor is observed as a light track marked by droplets in the Wilson fog chamber. The collisions between the elementary particles and the molecules of the vapor produce ions which cause the metastable vapor to decay locally into a two-phase system. The path is visualized analogously in a bubble chamber as a string of bubbles in a superheated liquid (e.g., liquid hydrogen at about 27 K).

4. Metastable states can also be of technical importance:

Designing a steam turbine, one has to know where the metastable state of steam breaks down into a stable wet steam state during the expansion. The region where this takes place is called Wilson-region.

The delay in the boiling of a liquid, the superheating, can destroy equipment by the following explosive evaporation of a part of the liquid. This can be prevented by introducing boiling chips into the liquid which hold small bubbles of unsolved gas or vapor at their rough surface.

The non-adiabatic transition of a system from a metastable phase into one stable phase can be observed at the surface of an air plane which flies through a cloud of subcooled water droplets. During the impact at the surface, the droplets do not only freeze partially. They deliver heat into the air and into the material of the air plane and freeze totally and form ice layers which caused many losses of planes.

At the windshield of a car which was parked for a longer time one can observe that subcooled droplets which are distributed by the wiper on the glass surface freeze immediately. The heat of freezing in this case is mainly delivered to the glass.

Sub-cooled water:

Freezing only in presence of „ice-crystal embryos“

-5°C 50000 water molecules for stability

-20°C a few hundred water molecules for stability

-40°C 70 water molecules for stability

Homogeneous ice formation practically only at -40°C

More examples for metastable states:

Diamond

tin pest

injection of silver iodide (AgI) into humid air from planes, to enforce rain

„snow guns“

3.6 Linde-Process for the Liquefaction of Gases

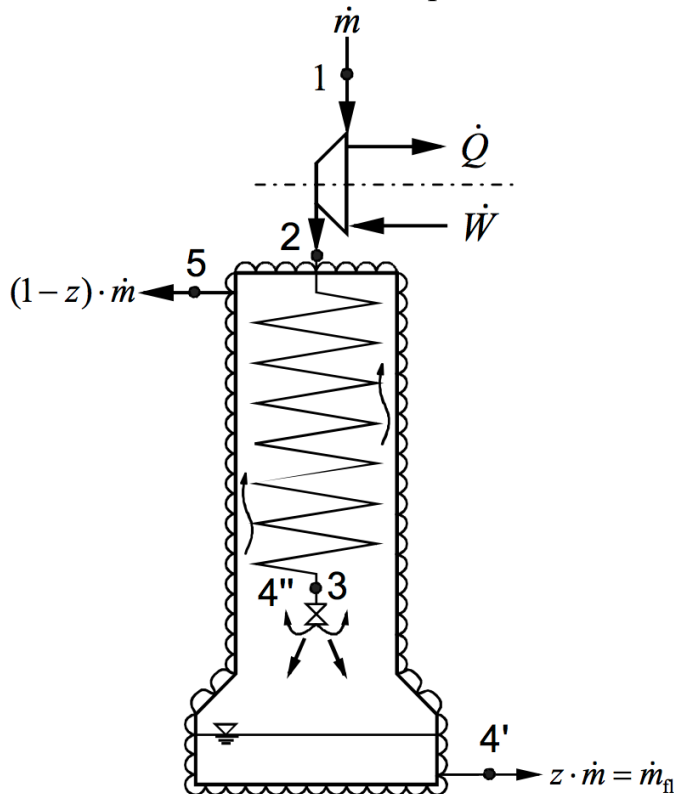


Figure: Linde-process, scheme of apparatus

1 → 2 isothermal compression to about 200 bar (in practice: compression, then cooling)

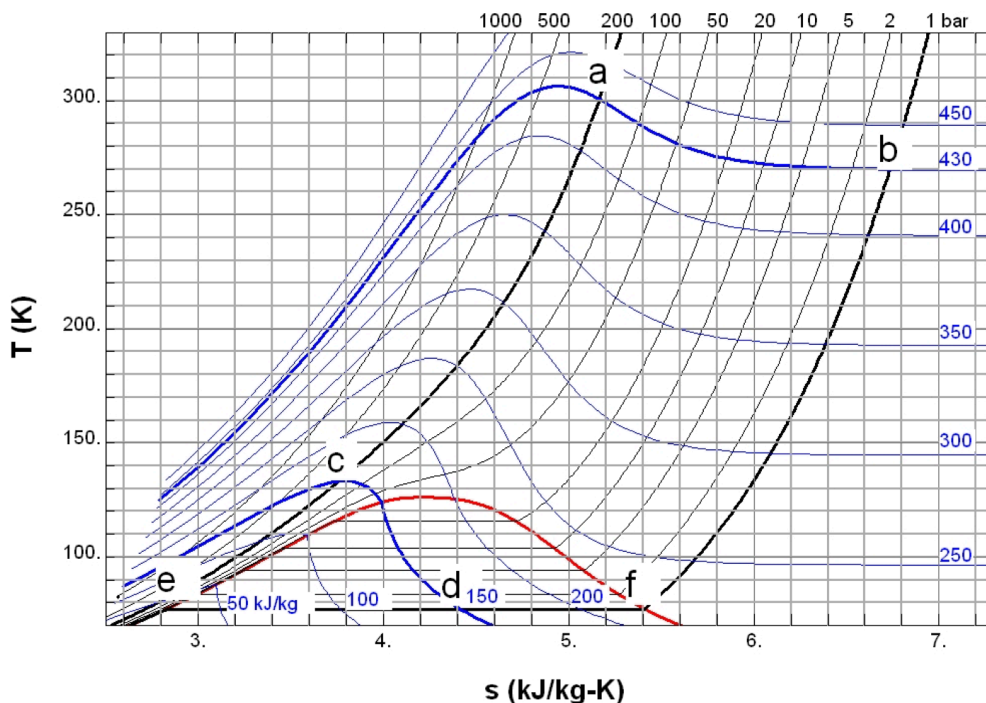
2 → 3 cooling of the compressed air in the thermally isolated counter flow heat exchanger

3 → 4 expansion of the air by throttling; the produced liquid air 4' is removed; the remaining gaseous air 4'' flows through the counter-flow heat exchanger where it cools the incoming compressed air

4'' → 5 heating of the air in the counter-flow heat exchanger to nearly ambient temperature

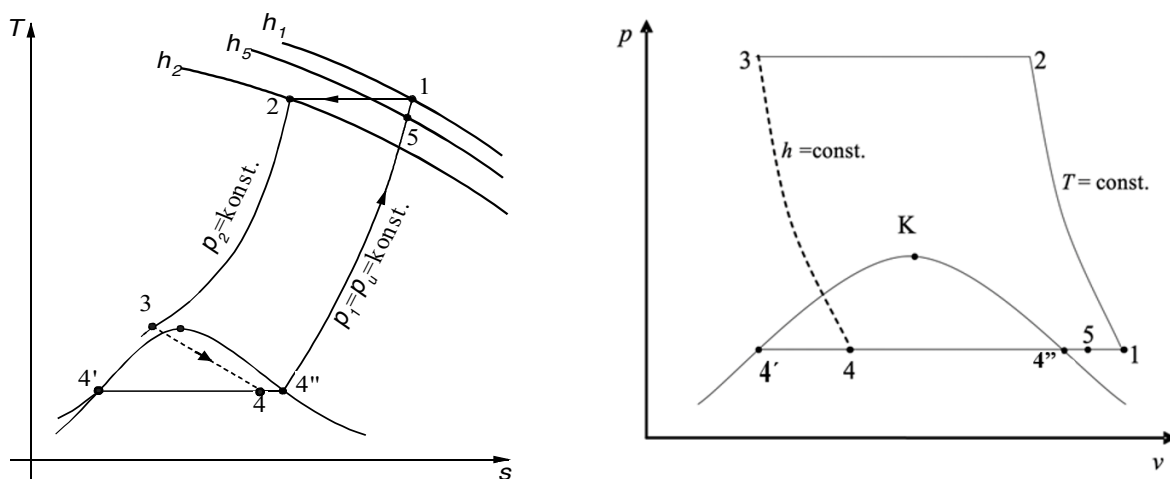
Historical notes

The Joule-Thomson effect is named after James Prescott Joule and Sir William Thomson (who later was promoted to Lord Kelvin), who described the phenomenon in 1852.



Inversion of the Joule-Thomson effect, depicted in a T, s -diagram.

Source: https://en.wikipedia.org/wiki/File:Throttling_in_Ts_diagram_01.jpg



Idealized Linde-Process in a T, s and in a p, v -diagram.

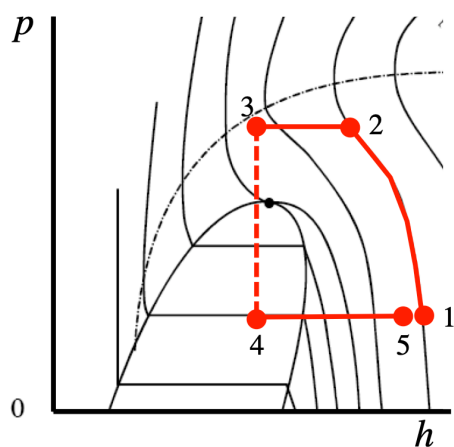


Figure: Idealized Linde-process in a p, h -diagram

Condition for a temperature decrease from 3 to 4 outside of the liquid-vapor two-phase region (cf. T, s -diagram): System is below inversion curve

Operation parameters:

$$T_1 \approx T_a ; p_1 \approx p_a ; p_2 \approx 20 \text{ MPa} = 200 \text{ bar}$$

Boiling temperatures at $p = 0.1 \text{ MPa}$

$$O_2 : -183^\circ\text{C} ; N_2 : -196^\circ\text{C} ; \text{air (averaged)} : -194^\circ\text{C}$$

Critical temperatures:

$$O_2 : -119^\circ\text{C} ; N_2 : -147^\circ\text{C} ; \text{air} : -141^\circ\text{C}$$

Air as a mixture of O_2 and N_2 has no constant boiling temperature at constant pressure. Its critical behavior is also different from that of a pure substance. The mentioned values of the temperature should only provide a rough idea.

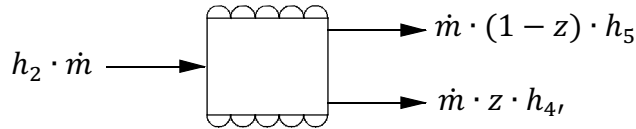
How large is the yield of liquid air?

Mass balance: $\dot{m}$: mass flow in the heat exchanger

$\dot{m}_l$: mass flow of liquid air

$z = \frac{\dot{m}_l}{\dot{m}}$ $\dot{m} - \dot{m}_l$: mass flow of gaseous air out of the heat exchanger

Enthalpy balance for the counterflow heat exchanger:



$$\dot{m} \cdot h_2 - z \cdot \dot{m} \cdot h_{4'} - (1 - z) \cdot \dot{m} \cdot h_5 = 0$$

$$z \cdot (h_5 - h_{4'}) = h_5 - h_2;$$

$$z = \frac{h_5 - h_2}{h_5 - h_{4'}}$$

z is the yield (fractional part of the air that is liquefied).

What if there was ideal gas behavior?

$$\rightarrow T_5 < T_2$$

$\rightarrow h_5$ would be smaller than $h_2 \rightarrow$ process would be impossible

Liquid air has a lower enthalpy compared to the incoming air. The cooling will work only if the outgoing (gaseous) air has a higher enthalpy than the incoming air.

The classical Linde-process with its throttling of the total air flow and the very high pressures (~20 MPa) is now about 100 years old. It was early modified to the Claude process. In this process a partial flow of the air is expanded in a turbine. The work from this expansion is an additional cooling effect which causes a higher yield of liquid air. Such expansions are used in large plants. Also, the maximum pressure is much lower today (~2.5 MPa). Nevertheless, the classical Linde process and the Claude process are still interesting on the way to the understanding of large modern plants for the liquefaction of gases.

Modification of the Linde Process: Claude Process

Difference to the Linde process: A partial flow $\dot{m} \cdot z^*$ is expanded in a turbine from p_2 to p_a .

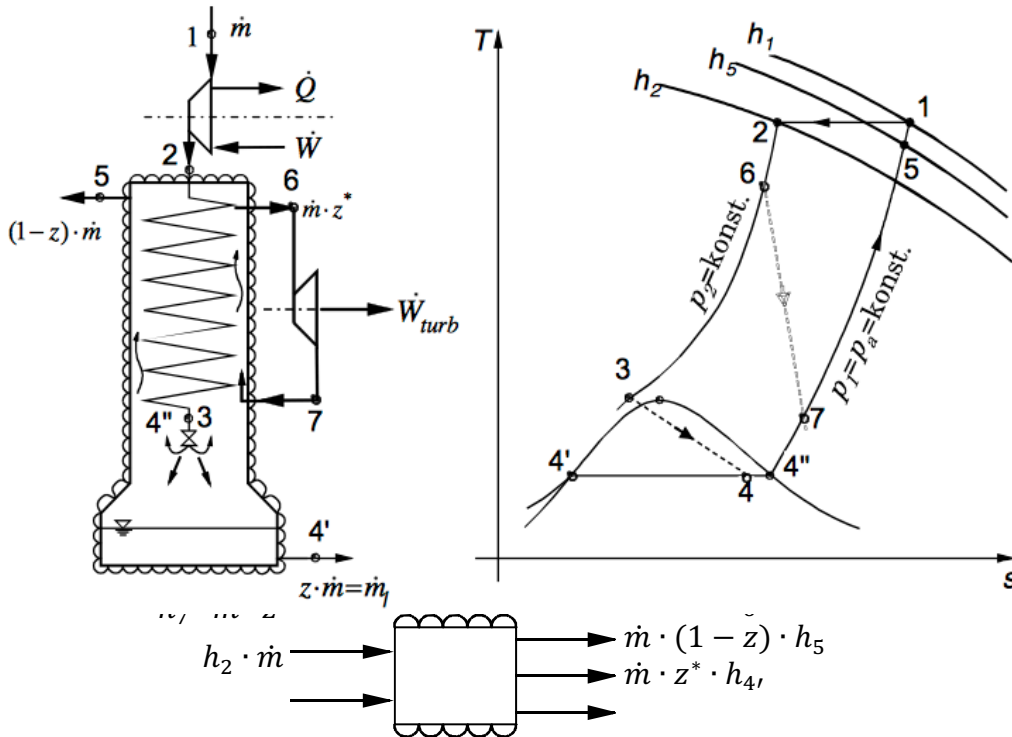


Figure 9.25: Claude-process

1 → 5 the same as in the Linde process

6 → 7 expansion of the partial flow $\dot{m} \cdot z^*$ in the turbine

Enthalpy balance for the counter flow heat exchanger:

$$\dot{m} \cdot h_2 - z \cdot \dot{m} \cdot h_{4'} - (1 - z) \cdot \dot{m} \cdot h_5 - z^* \cdot \dot{m} \cdot (h_6 - h_7) = 0$$

$$z = \underbrace{\frac{h_5 - h_2}{h_5 - h_{4'}}}_{z_{\text{Linde}}} + z^* \cdot \frac{h_6 - h_7}{h_5 - h_{4'}}$$

As mentioned above, the decrease of the enthalpy of the partial flow $\dot{m} \cdot z^*$ in the turbine from h_6 to h_7 corresponds to an additional *cooling* and causes an increase of the yield z . The work delivered by the turbine is of minor interest.

Decomposition of air

Boiling points:

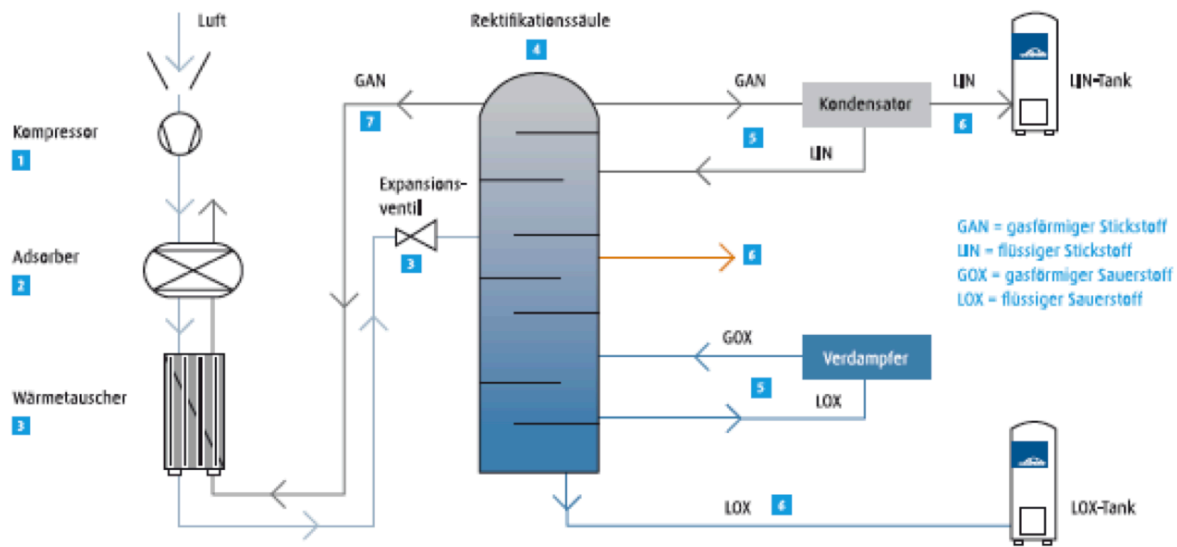
Oxygen 90.2 K (-183 °C)

Argon 87.3 K (-186 °C)

Nitrogen 77.4 K (-196 °C)

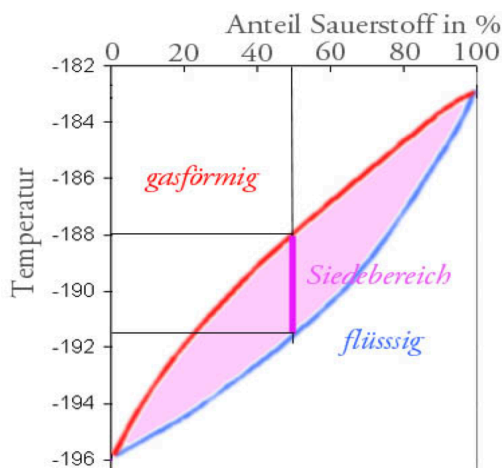
Die Luftzerlegung im Überblick

- | | | | |
|----------------------------------|--------------------------------------|----------------------------|---------------------------------|
| 1 Verdichtung | 2 Vorkühlung und Reinigung | 3 Verflüssigung | 4 Luftzerlegung (Rektifikation) |
| 5 Wiederholung der Rektifikation | 6 Lagerung – Gewinnung von Edelgasen | 7 Stickstoff-Ausschleusung | |



Source: Linde

Evaporation behavior of N_2/O_2 mixtures (Source: <http://de.academic.ru/pictures/dewiki/83/Sdp7.png>)



Applications of liquid nitrogen:

- Preservation of biological substances
- Freezing of water pipes
- Cryotherapy
- Cooling traps
- Overclocking of CPUs
- Freezing of soil

Applications of liquid oxygen:

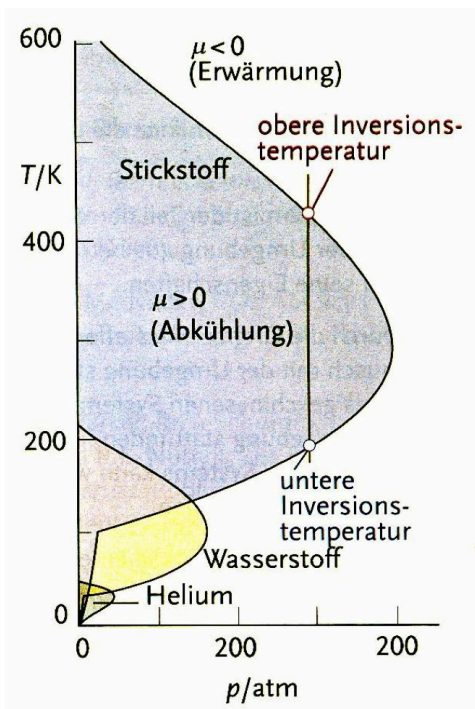
- Processing of raw steel
- Process engineering
- Cutting of iron
- Medical applications
- Hot flames (welding, glass production)
- Rocket fuels (LOX)

Applications of Argon:

- Inert gas (e.g., welding of titanium, tungsten)
- Process engineering
- Lamps, gas discharge lamps

Liquefaction of Hydrogen and Helium

Pre-cooling below the inversion temperature



Source: P. Atkins, "Physikalische Chemie" (English edition title: Physical Chemistry)